

Resource Provisioning Analysis for Large-Scale GStreamer MCU Video Mixing

Technical Capacity Assessment

July 2026

1 Executive Summary

This analysis evaluates the hardware capacity for **1,000 concurrent MCU rooms** using **GStreamer** pipelines in MCU mode (real-time decoding, compositing, scaling, and encoding of multiple 720p streams into single 1080p outputs), with GPU acceleration (NVIDIA NVENC/CUDA) and lower DDR4 memory configurations.

- 80% (800 rooms): 6-participant rooms
- 20% (200 rooms): 20-participant rooms

The proposed cluster consists of **40 nodes** (20 Small + 20 Big) and comfortably supports the workload with headroom.

2 Node Specifications

Table 1: Node Types and Specifications (GPU-Accelerated, Lower DDR4)

Component	Small Node (6-person)	Big Node (20-person)
Target Use	6-participant rooms	20-participant rooms
CPU	Single Xeon Silver/Gold (16–24 cores)	Dual Xeon Gold/Platinum (32–64+ cores)
RAM	64–96 GB DDR4	128 GB DDR4
GPU	1 × Mid-High NVIDIA (e.g., RTX 4090 / A10 class)	2 × High-End NVIDIA (e.g., A40 / L40S class)
Network	25–40 Gbps	40–100 Gbps
Storage	256 GB NVMe SSD	512 GB NVMe SSD

Table 2: Real-Time Capacity per Node

Metric	Small Node	Big Node
Rooms per Node	45	12
Input Streams per Node	≈ 270	≈ 240
Output Streams per Node	45	12

3 Per-Node Capacity (GStreamer MCU)

4 Cluster Capacity with 20 Small + 20 Big Nodes

Table 3: Total Cluster Capacity

Category	Small Nodes (20)	Big Nodes (20)	Total / Headroom
6-Person Rooms	900	-	900 (+100 headroom)
20-Person Rooms	-	240	240 (+40 headroom)
Max Mixed Rooms	-	-	1,140
Target Workload	800	200	1,000 (covered with 14% headroom)

5 Resource Totals

Table 4: Cluster-Wide Resource Summary

Resource	Total
Total Nodes	40
Total DDR4 RAM	$\approx 3.8\text{--}4.2\text{ TB}$
Total NVIDIA GPUs	60
Estimated Peak Power	High (GPU-heavy)

6 Conclusions and Recommendations

The proposed 40-node heterogeneous cluster provides robust capacity for 1,000 concurrent GStreamer MCU rooms under GPU acceleration. Headroom supports maintenance, traffic spikes, and additional features (e.g., recording).

Future work may include dynamic auto-scaling, detailed pipeline profiling with `gst-debug`, and power-efficiency analysis.